

6 The matrices **M** and **N** are $\begin{pmatrix} \lambda & 2 \\ 2 & \lambda \end{pmatrix}$ and $\begin{pmatrix} \mu & 1 \\ 1 & \mu \end{pmatrix}$ respectively, where λ and μ are constants.

(a) Investigate whether **M** and **N** are commutative under multiplication. **[2]**

(b) You are now given that $\mathbf{MN} = \mathbf{I}$.

(i) Write down a relationship between $\det \mathbf{M}$ and $\det \mathbf{N}$. **[1]**

(ii) Given that $\lambda > 0$, find the exact values of λ and μ . **[3]**

(iii) Hence verify your answer to part **(i)**. **[2]**